

VISION_NR_Temporal

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Overview

- Previously, we discussed classical methods for noise reduction - VISION_NoiseReduction.
- There, patch based methods performed the best.
- Then, we discussed ml based methods for noise reduction - VISION_NoiseReduction_ML.
- There, a UNet architecture, residual learning, and infusing noise information (noise maps, and camera ISO, exposure time, f#) into the learning process, produced the best results.

Video

- For video, spatial based denoising can be applied to each and every video frame.
- Unfortunately, temporal artifacts like flickering, popping, etc. often result, due to the lack of temporal coherency.
- To mitigate these artifacts, classical denoising algorithms like Non Local Means and BM3D extend their patch based methods to neighboring video frames.
- Another approach uses computational photography, pixel alignment, and temporal averaging.

Perfect Alignment and Raw Data Frames

- For a static scene, N consecutive raw video frames are captured and averaged.
- This produces the following per pixel noise reduction: $\frac{\sigma^2}{N}$
- In VISION_NoiseReduction_ML, we discussed how this result served as a “best case scenario” for temporal noise reduction.
- i.e. The “best case scenario” degrades in the presence of correlated noise, biased noise, etc.
- Assuming the best case scenario, different practical matters need to be addressed.

In Practice

- First, the capture may be handheld.
- If true, some type of stabilization may be applied, using a lens shift or sensor shift.
- Next, the captured frames need to undergo global registration (rotational and translational) to a reference frame.
- In practice, different similarity transforms are computed and applied.

In Practice

- Rotational information can be obtained from the internal camera gyro.
- Global translational shifts can be computed in software - akin to electronic image stabilization.
- The neighboring image frames can then be warped to a reference frame.
- In a perfect world -say a static landscape scene where everything is at infinity and all warping parameters are computed with absolute accuracy- alignment will be perfect.

In Practice

- In practice, there will be small errors.
- In practice, the scene will usually be significantly more complex.
- For example, the scene may have depth - static trees / moving cars / people walking, in both the foreground and the distant background.
- This creates a parallax problem, and the subsequent need for local alignment.

In Practice

- Optical flow handles parallax well.
- For implementation, optical flow can be performed in a coarse to fine manner, using a pyramid structure.
- This helps performance, in high noise environments.
- Optical flow degrades, when brightness changes occur between neighboring frames.
- Optical flow degrades in textureless regions of a scene, such as the sky or walls.

Weight Map / Confidence Map

- In a perfect world, the previous steps would produce perfect alignment - however, the world is not perfect.
- If strict averaging is applied in the “imperfect world”, new temporal artifacts will be created.
- To mitigate this problem, a weight map / confidence map needs to be computed and applied at the pixel level.
- If the confidence is high, the pixel for the reference frame and the pixel for the neighboring frame, will have similar weight.
- If the confidence is low, the neighboring frame pixel will be significantly de-weighted.

The Math

- In practice, we compute the difference in brightness between the reference and the (warped) neighboring frame at every pixel location.

$$diff_{bright}(x_i, y_j) = |I_{ref}(x_i, y_j) - I_{neigh}(x_i, y_j)|$$

$$diff_{brightnorm}(x_i, y_j) = \frac{|I_{ref}(x_i, y_j) - I_{neigh}(x_i, y_j)|}{\epsilon + |I_{ref}(x_i, y_j)|} \text{ or } \frac{|I_{ref}(x_i, y_j) - I_{neigh}(x_i, y_j)|}{\sigma(x_i, y_j)}$$

$$w(x_i, y_j) = \exp\left(-\frac{diff_{brightnorm}(x_i, y_j)}{\tau}\right) \text{ or } w(x_i, y_j) = \frac{1}{1 + \left(\frac{diff_{brightnorm}(x_i, y_j)}{\tau}\right)^2}$$

$$w(x_i, y_j) = \text{clamp}\left(1 - \frac{diff_{bright}(x_i, y_j)}{k\sigma(x_i, y_j)}, [0, 1]\right)$$

$$I_{new}(x_i, y_j) = w(x_i, y_j)I_{ref}(x_i, y_j) + (1 - w(x_i, y_j))I_{neigh}(x_i, y_j)$$

When Brightness Fails

- Using the brightness difference works well, assuming the frame to frame exposure / brightness is fixed.
- However, situations can arise, where the brightness changes between frames, but the underlying scene structure does not.
- In these cases, the brightness based difference and weighting map will be ineffective at temporal denoising.
- In these situations, a gradient based difference and weighting map may prove helpful.

The Math

- In practice, we compute the difference in gradient direction between the reference and the (warped) neighboring frame at every pixel location.

$$g_{ref} = \frac{\nabla I_{ref}}{\max(||\nabla I_{ref}||, \epsilon)} \quad \text{and} \quad g_{neigh} = \frac{\nabla I_{neigh}}{\max(||\nabla I_{neigh}||, \epsilon)}$$

$$\nabla I_{neigh} = g \nabla I_{ref}$$

$$w(x_i, y_j) = \cos(\theta) \approx \frac{\nabla I_{ref} \cdot \nabla I_{neigh}}{||\nabla I_{ref}|| ||\nabla I_{neigh}|| + \epsilon} \rightarrow \frac{g ||\nabla I_{ref}||^2}{g ||\nabla I_{ref}||^2 + \epsilon} \approx 1$$

- Notice how the weight is (now) independent of g.
- If the cosine similarity is close to 0 (different directional gradients), no denoising will occur.

In Practice

- Improved temporal denoising robustness can result from computing both the brightness and gradient based weight maps.
- Note: ***The gradient based approach is still effective under local gain variations, as long as the gain variations are smooth.
- Caveats
- The gradient approach can alter the temporal look and feel, since it computes an average using a term with a gain offset.
- In practice, if the gain is uniform, normalizing the image intensities might provide the better user experience.
- ***If the gain results from changes in tone mapping (and the temporal denoising is applied post tone mapping), pumping / breathing can result if the denoising tracks the tone mapping changes.

Problematic Regions

- Textureless regions - be it sky or dark walls, are problematic areas that affect motion estimation.
- Low contrast textured regions in low light are problematic for gradient operations, because they are hard to discern / tease apart from image noise.
- Finally, fine structures like human hair are problematic, since small alignment errors will quickly smooth the underlying structure.

Hair

- Two approaches can be taken - an object centric approach or a frequency centric approach.
- AI detectors can perform object detection and segmentation.
- However, errors may exist, creating a new set of temporal artifacts.
- Traditional signal process is agnostic to the object, but sensitive to the frequency content.
- Used in a pyramid structure, good results can be obtained.

Hair

Procedure

- Warp the neighboring frames to the reference frame, in the usual manner.
- Then, decompose the reference and neighboring frames into two separate images - a low frequency image and a high frequency image.
- Perform aggressive temporal averaging on the low pass filtered images.
- Perform weak temporal averaging (and moderate spatial averaging) on the high pass filtered images.
- Recombine the new low pass and high pass image components.

Hair

Example:

$$I_{ref}(x, y) \text{ and } I_{neigh}(x, y)$$

$$\tilde{I}_{neigh}(x, y) = \text{warp}(I_{neigh}(x, y))$$

$$I_{ref}^{low}(x, y) = \text{Blur}(I_{ref}(x, y)) \text{ and } I_{ref}^{high}(x, y) = I_{ref}(x, y) - I_{ref}^{low}(x, y)$$

$$\tilde{I}_{neigh}^{low}(x, y) = \text{Blur}(\tilde{I}_{neigh}(x, y)) \text{ and } \tilde{I}_{neigh}^{high}(x, y) = \tilde{I}_{neigh}(x, y) - \tilde{I}_{neigh}^{low}(x, y)$$

$$w^{low}(x, y) = 0.5 \text{ and } w^{high}(x, y) = 0.9$$

$$\hat{I}^{low}(x, y) = w^{low} I_{ref}^{low}(x, y) + (1 - w^{low}) \tilde{I}_{neigh}^{low}(x, y)$$

$$\hat{I}^{high}(x, y) = w^{high} I_{ref}^{high}(x, y) + (1 - w^{high}) \tilde{I}_{neigh}^{high}(x, y)$$

$$\rightarrow \hat{I}(x, y) = \hat{I}^{low}(x, y) + \hat{I}^{high}(x, y)$$

Hair

- This procedure can proceed “up” a pyramid structure (L0, L1, L2,)
- Now, the weights for the low resolution pyramid levels will be higher than the weights for the high resolution levels.
- Noise reduction results, while preserving high frequency structure.

Summary

- A slide presentation was generated, focusing solely on temporal noise reduction.
- The temporal noise reduction approach was rooted in pixel alignment / computational photography.
- A weighting scheme was introduced to address regions with “non perfect” pixel alignment, enabling regression to the reference image.
- For very high frequency structures not amenable to small pixel alignment errors, AI based segmentation, or, signal processing based frequency band decomposition was suggested.
- Signal processing based “signal decomposition using pyramids”, is a powerful tool from both performance and computational standpoints.